

Negative Literature Answer to the $V(T)$ -Geometry Question in arXiv:1706.08682

FA/Banach run packet

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Status

This is a `literature_already_answered` packet. It records that the open question in Hernández–Semenov–Tradacete [1] was answered negatively by the same authors in the later paper [2]. No novelty is claimed for this note.

Original question

For an operator $T : L_\infty \rightarrow L_1$, set

$$V(T) = \left\{ \left(\frac{1}{p}, \frac{1}{q} \right) : T : L_p \rightarrow L_q \text{ is strictly singular and non-compact} \right\}.$$

The examples in [1] had $V(T)$ assembled from line segments that were horizontal, vertical, or parallel to the diagonal. Immediately after the displayed example, the authors asked whether $V(T)$ must always look like that.

Answer

The later paper [2] explicitly restates this question and announces a negative answer. Its main realization theorem gives considerably more.

Let $\mathfrak{L}$ be the family of line segments $\ell \subset (0, 1)^2$ such that either

1. ℓ is horizontal or vertical and avoids the central forbidden region specified in [2]; or
2. ℓ has positive slope and lies entirely below the diagonal.

For such a segment, let L_ℓ be the side of the square selected in [2].

Literature Answer (Hernández–Semenov–Tradacete, Theorem 9). *For every $\ell \in \mathfrak{L}$, there is an operator $T : L_\infty \rightarrow L_1$ such that*

$$L(T) = S(T) = L_\ell \cup \ell, \quad K(T) = L_\ell, \quad V(T) = \ell.$$

Consequently, the question in [1] has a negative answer.

Indeed, choose an admissible segment below the diagonal of slope 2 (or any positive slope other than 1). The theorem produces an operator with $V(T) = \ell$. This segment is neither horizontal, vertical, nor parallel to the diagonal, so it cannot have the geometry suggested by the 2017 example.

Construction and proof idea

For $0 < \alpha, \lambda < 1$, let $\Omega_\alpha \subset (0, 1)$ be an Ahlfors α -regular set and consider the Riesz potential

$$(T_\lambda f)(t) = \int_0^1 \frac{f(u)}{|t-u|^\lambda} du, \quad t \in \Omega_\alpha.$$

The supporting paper computes its V -set exactly:

$$V(T_\lambda) = \left\{ \left(x, \frac{x-1+\lambda}{\alpha} \right) : 1-\lambda < x < \min \left\{ 1, \frac{1-\lambda}{1-\alpha} \right\} \right\}.$$

This is a segment of slope $1/\alpha > 1$. The proof combines weak-type estimates, interpolation, strict singularity of positive integral operators off the diagonal, and a concentration sequence proving non-compactness. Duality supplies slopes between 0 and 1. Separate factorizations through Rademacher and disjoint sequences produce the horizontal and vertical cases.

Verification and scope

The source PDF poses the question on page 14. The supporting PDF identifies that precise question on page 2 and states the exact realization theorem on page 8. Both papers use the same $V(T) = S(T) \setminus K(T)$ definition for operators $L_\infty \rightarrow L_1$. The existing run packet about arXiv:2001.09677 addresses a different endpoint question in that later paper, so it is not a duplicate of this result.

References

- [1] F. L. Hernández, E. M. Semenov, and P. Tradacete, *Interpolation and extrapolation of strictly singular operators between L_p spaces*, Adv. Math. 316 (2017), 667–690; arXiv:1706.08682.
- [2] F. L. Hernández, E. M. Semenov, and P. Tradacete, *Strictly singular non-compact operators between L_p spaces*, Rev. Mat. Iberoam. 39 (2023), no. 1, 181–200; arXiv:2001.09677; doi:10.4171/RMI/1360.